

Data Flow Diagram

Date	06-07-2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Data Flow Diagram

The diagram below shows how data moves through the OptiCrop application between external entities, processes, and data stores, based strictly on the implemented Flask backend, machine learning prediction pipeline, and database layer. Use the symbol legend below as a guide.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	External Entity	A person, organization, or system outside the project that sends or receives data (e.g., Customer, Payment Gateway).
Rectangle with numbered header	Process	An activity that transforms incoming data into outgoing data (e.g., Validate Request, Process Order).
Rectangle (solid fill, no number)	Data Store	A place where data is held for later use (e.g., Customer Database, Inventory).
Labeled arrow	Data Flow	The movement of data between entities, processes, and data stores, labeled with what data is being passed.

OptiCrop Data Flow Diagram

The diagram on the following page illustrates the verified data flow of the OptiCrop system: user-submitted soil and climate parameters, input validation, ML-based crop prediction using the trained model artifacts, crop reference lookups, storage and retrieval of prediction records in the SQLite database, dashboard analytics, CSV history export, and PDF report generation.

OptiCrop: Smart Agricultural Production Optimization Engine — Data Flow Diagram

